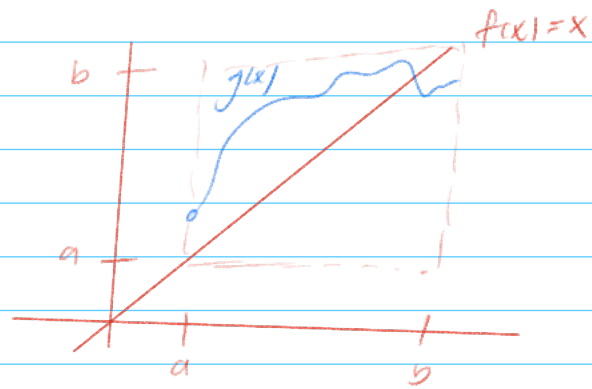
Thm Existence & uniqueness of Fixed Pts

Assmp: $g \in C[a,b]$; $g(x)$ is a contraction mapping i.e.
$g([a,b]) \subset [a,b]$

$\exists x^* \in [a,b]$ s.t. $g(x^*) = x^*$

intuit:



$\exists$ pf: If $g(a) = a$ or $g(b) = b$, then g has fixed pts @ a or b.

Otherwise $g(a) > a$, $g(b) < b$.

let $h(x) := g(x) - x \implies h(a) > 0$, $h(b) < 0$

Note $h \in C[a,b]$ $\therefore$ By IVT $\exists x^* \in [a,b]$ s.t.

$h(x^*) = 0 \implies \exists x^*$ s.t. $g(x^*) - x^* = 0 \implies g(x^*) = x^*$

end of existence part

$\exists!$ pf: STAC $\exists x^*, y^* \in [a,b]$ $x^* \neq y$ and
$g(x^*) = x^*$, $g(y^*) = y^*$

extra assumption of $C'[a,b]$

MVT: $f \in C[a,b]$ and $f \in C'(a,b)$ $\exists \xi \in (a,b)$ s.t.

$$f(\xi) = \frac{f(b) - f(a)}{b - a}$$

$\Rightarrow \exists \xi \in (x^*, y^*)$ s.t. $g'(\xi) = \dfrac{g(x^*) - g(y^*)}{x^* - y^*} = 1$

However $\xi \in (x^*, y^*) \subseteq (a,b) \Rightarrow |g'(\xi)| \le k < 1 \Rightarrow\Leftarrow$

$\Rightarrow x^* = y^*$ $\therefore$ x^* is a unique fixed pt.

Fixed-Pt Iteration

Algorithm: finding x^* s.t. $g(x^*) = x^*$

1.) Choose init. guess x_0
2.) Generate $\{x_n\}_{n=1}^{\infty}$ by letting $x_n = g(x_{n-1})$ for $n \ge 1$. If $x_n \to x^*$ as $n \to \infty$ and $g(x)$ is cont., then

$$x^* = \lim_{n\to\infty} x_n = \lim_{n\to\infty} g(x_{n-1}) = g(\lim_{n\to\infty} x_{n-1}) = g(x^*)$$

<u>Thm</u>: let $g(x) \in C[a,b]$ s.t. $g(x) \in [a,b]$

<u>Error bound</u>

$|x_n - x^*|$

Copy from online!

Newton's Method

Root finding problem, i.e. f s.t. $f(x) = 0$

Idea: convert to fixed-pt problem $\to$ local linear approx.

<u>Algo</u> (Newton's Metn): $f \in C^2[a,b]$, find $f(x^*) = 0$

<u>Step 1</u>: Choose an initial guess x_0
<u>Step 2</u>: For $n \ge 1$

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}, \quad \text{for } n \geq 1$$

<u>Step 3</u>: If stopping crit. is met stop; else $n \leftarrow n+1$

<u>Stopping criterion</u>

- $|x_n - x_{n-1}| < \varepsilon$
- $\dfrac{|x_n - x_{n-1}|}{|x_n|} < \varepsilon$ where $p_n \neq 0$
- $|f(x_n)| < \varepsilon$

<u>Intuit</u>: $f \in C^2[a,b]$, $f(x^*) = 0$, init guess $x_0 \in [a,b]$ s.t.

① $f'(x_0) \neq 0$
② x_0 <u>close to</u> x^*

By Taylor Expansion:

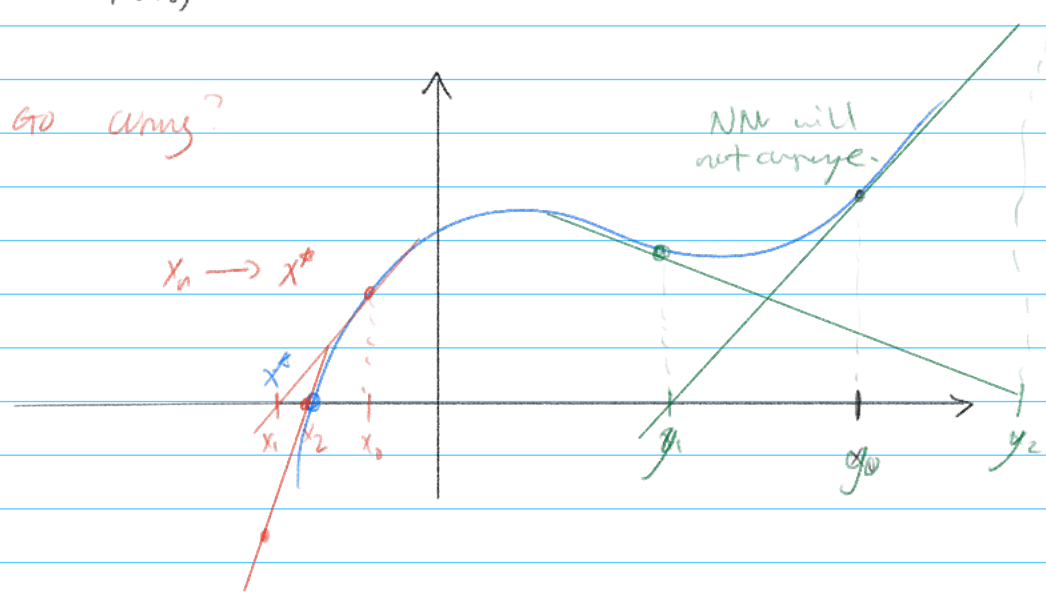
$$f(x^*) = f(x_0) + f'(x_0)(x^* - x_0) + o(|x^* - x_0|)$$

$$\Rightarrow \quad 0 \approx f(x_0) + f'(x_0)(x^* - x_0)$$

$$\Rightarrow \quad x^* \approx x_0 - \frac{f(x_0)}{f'(x_0)} := x_1$$

Is $x_1 \in [a,b]$?

What could go wrong?



Ex: $f(x) = x^2 - 3 \qquad f'(x) = 2x$

$$x_n = x_{n-1} - \frac{x_{n-1}^2 - 3}{2x_{n-1}}$$

$x_0 = 1.5$

$x_1 = 1.75$

$x_2 = 1.7321$

$x_3 = 1.7321$

Why is f C^2?

Thm [Convergence of Newton's Method] Let $x^* \in [a,b]$ be a zero of $f \in C^2[a,b]$. If $f'(x^*) \neq 0$ and $f''(x^*) < \infty$ then $\exists \delta > 0$, s.t. $\forall$ initial guess $x_0 \in (x^* - \delta, x^* + \delta)$, the seq. $\{x_n\}$ generated by Newton's method converges quadratically to x^* i.e. converges with order 2.

2 parts

Given: $f \in C^2[a,b]$, $x^* \in [a,b]$ s.t. $f(x^*) = 0$. $f'(x^*) \neq 0$, $f''(x^*) < \infty$

$\Rightarrow$ (1) $\exists \delta > 0$ s.t. $\forall x_0 \in (x^* - \delta, x^* + \delta)$ $x_n \xrightarrow{NM} x^*$

pf: let $\phi(x) = x - \frac{f(x)}{f'(x)}$ iteration function

$\phi(x) = x - \frac{f(x)}{f'(x)}$

N.M can be written as

$$x_{k+1} = \phi(x_k) \qquad \text{Fixed-pt finding}$$

and $x^* = \phi(x^*)$ WTS $x_k \to x^*$

let $e_k := x_k - x^*$ (error) $\qquad e_k \to 0 \Leftrightarrow x_k \to x^*$

Note $e_{k+1} = x_{k+1} - x^* = \phi(x_k) - x^* = \phi(x_k) - \phi(x^*)$

By MVT $\exists \xi \in (x_k, x^*)$ s.t. $e_{k+1} = \phi'(\xi)(x_k - x^*)$

$$\Rightarrow e_{k+1} = \phi'(\xi) e_k \quad (1), \quad \phi'(x) = 1 - \frac{f'^2 - f''f}{(f')^2}$$

$$= \cancel{1 - 1} + \frac{f''f}{(f')^2}$$

$$e_{k+1} = \phi'(\xi)\, e_k = \frac{f''(\xi) f(\xi)}{(f'(\xi))^2}$$

$f'(x^*) \neq 0$ ξ

$f''(x^*) < \infty$

$$\phi'(x^*) = \frac{f(x^*) f''(x^*)}{f'(x^*)^2} = 0 \quad (2)$$

$\phi'(x) \in C[a,b]$

$\Rightarrow \exists \delta > 0$, s.t. $x \in (x^* - \delta, x^* + \delta)$ s.t. $|\phi'(x)| \leq \frac{1}{2}$

From (1) $e_{k+1} = \phi'(\xi_k)\, e_k$ and $\xi_k \in (x_k, x^*)$

$\Rightarrow \xi_k \in (x^* - \delta, x^* + \delta)$

$\therefore |\phi'(\xi_k)| \leq 1/2 < 1$

$\Rightarrow |e_{k+1}| \leq \frac{1}{2} |e_k| = \delta$

$|e_{k+1}| \leq \frac{1}{2} |e_k| \leq \frac{1}{4} |e_{k-1}| \leq \dots \leq \frac{1}{2^{k+1}} |e_0| = \frac{1}{2^{k+1}} |x_0 - x^*|$

$\therefore |e_k| \to 0 \iff e_k \to 0$ $\cancel{\Delta}$